Advanced Programming Practice · 21CSC203P

Unit 1 · Session 11 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 11 : SLO – 1 | High-level to Machine Code

Activity (Solutions)

• 1) Compile + disassemble: javac Hello.java produces Hello.class; javap -c Hello shows the bytecode, e.g., main: 0: getstatic #2 (System.out) 3: ldc #3 ("Hello") 5: invokevirtual #4 (println) 8: return — mnemonics the JVM executes.

• 2) JVM bytecode → machine code: the interpreter executes bytecode instruction by instruction; the JIT (Just-In-Time) compiler detects hot spots (frequently run code) and compiles them to native machine code, which the CPU then executes directly.

• 3) Class loaders: Bootstrap loader loads core JDK classes (java.lang.*) from the runtime; Extension/Platform loader loads extension libraries; Application/System loader loads classes from the application classpath. They work in a delegation hierarchy — each loader first asks its parent, ensuring core classes cannot be overridden and classes load once.